

MARKETING CAMPAIGN ANALYTICS

Customer Segmentation & Campaign Effectiveness — Project Report

Domain: Marketing Analytics / Customer Analytics

Prepared using Python (Pandas, EDA), MySQL, and Power BI

Dataset: 56,000 customer records | 20+ variables across demographics, spending, channels, and campaigns

Executive Summary

This project analyzes 56,000 customer records for a retail company to understand who its most valuable and most responsive customers are, what drives marketing campaign acceptance, and how customer segments differ in spending and channel behaviour. The analysis combines Python-based data cleaning and exploratory data analysis, a MySQL data model with analytical queries and views, and an interactive two-page Power BI dashboard (Executive Dashboard and Customer Insights Dashboard).

The customer base generated ₹35.86M in total revenue at an average spend of ₹640.33 per customer, with an overall campaign response rate of 14.76%. Income emerged as the strongest driver of both spending and campaign response, followed by age and family size. Customers with higher income, fewer children, and moderate-to-occasional (rather than frequent) website engagement responded to campaigns at the highest rates, while a large under-served segment — 16,765 customers (29.9% of the base) — browses the website frequently but spends little and has not responded to any campaign.

Key headline numbers: 56,000 customers | ₹35.86M total revenue | ₹57.25K average income | ₹640.33 average spend | 14.76% overall response rate | 8,265 campaign responders.

Detailed findings, segment profiles, dashboard visuals, and 4-6 actionable recommendations for future campaigns are presented in the sections that follow.

1. Problem Statement

A retail company has run multiple marketing campaigns and recorded detailed information about its customers — demographics, spending across product categories, channel-wise purchases, and whether they accepted specific campaigns. Management wants a consolidated analytics solution to understand:

- Who its best (most valuable and most responsive) customers are.
- What drives campaign acceptance across different customer groups.
- How customer segments behave across products and purchase channels.

This report addresses the following business questions using the cleaned dataset, SQL-based KPIs, and the Power BI dashboard built for this project:

- Which customer segments have the highest response rate to campaigns, overall and per campaign?
- How do spending patterns across products vary by age, income, marital status, and country?
- Which channels are most used by high-value customers, and how often do they visit the website?
- Are there specific segments that are under-served — low spending, high website visits, low response?
- What are the characteristics of an "ideal target customer" for future campaigns?

2. Approach & Methodology

2.1 Data Understanding & Cleaning (Python)

- Loaded marketing_campaign_data.csv (56,000 rows) and cross-referenced the data dictionary file for field definitions.
- Verified there were no duplicate customers (row count matched unique ID count) and no null values across the dataset.

- Converted Dt_Customer to a proper date type and derived Customer_Tenure as days since the earliest reference date.
- Derived key features: Age (from Year_Birth), Total_Spend (sum of all product spend columns), Total_Purchases (sum of all channel purchase columns), and Children (Kidhome + Teenhome).
- Checked numeric fields (Age, Income, Total_Spend) for outliers using the IQR method; retained outliers found to represent genuine high-income / high-spending customers rather than data-entry errors.

2.2 Exploratory Data Analysis (Python)

- Univariate analysis of age, income, education, marital status, country, and recency distributions.
- Bivariate/multivariate analysis of spending vs. income, age, marital status, family size, and campaign response; channel usage vs. spending and response; web engagement vs. spending and response.
- Correlation analysis across numeric variables to identify the strongest drivers of spend and response.

2.3 Rule-Based Segmentation (Python & SQL)

- Implemented the six business-defined segments (High Income, Young Customer, Campaign Responder, High Web Engagement, Family Customer, High Spender) using conditional logic and percentile thresholds in Python.
- Re-created the same segment logic in SQL using CASE statements for validation and dashboard support.

2.4 SQL Data Modeling & Queries

- Created a MySQL database (Marketing_Campaign) with a primary customers table (ID as primary key) loaded from the cleaned CSV.
- Wrote analytical queries covering overall KPIs, demographic breakdowns, spending analysis, campaign analysis, channel analysis, and segment-level summaries.
- Created seven reusable SQL views (vw_country_revenue, vw_campaign_summary, vw_segment_summary, vw_kpi_summary, vw_product_summary, vw_channel_summary, vw_channel_totals) to feed the dashboard layer directly.

2.5 Dashboard (Power BI)

- Built a two-page Power BI report: an Executive Dashboard (top-line KPIs, revenue and campaign performance) and a Customer Insights Dashboard (segment-level behaviour), both with filters for Country, Education, Marital Status, Age, and Income.

3. Dataset Overview

The cleaned dataset covers 56,000 unique customers with no missing values or duplicate records. Key derived fields used throughout the analysis: Age, Total_Spend, Total_Purchases, Children, and Customer_Tenure.

Metric	Value
Total Customers	56,000
Total Revenue (Total_Spend)	₹35.86M
Average Income	₹57,252

Metric	Value
Average Spend per Customer	₹640.33
Overall Campaign Response Rate	14.76% (8,265 responders)
Age Range	30 – 90 years (mean 54.3, median 53)
Income Range	₹1,730 – ₹258,028 (mean ₹57,252, median ₹58,839)

4. Exploratory Data Analysis

4.1 Customer Demographics

Age Distribution

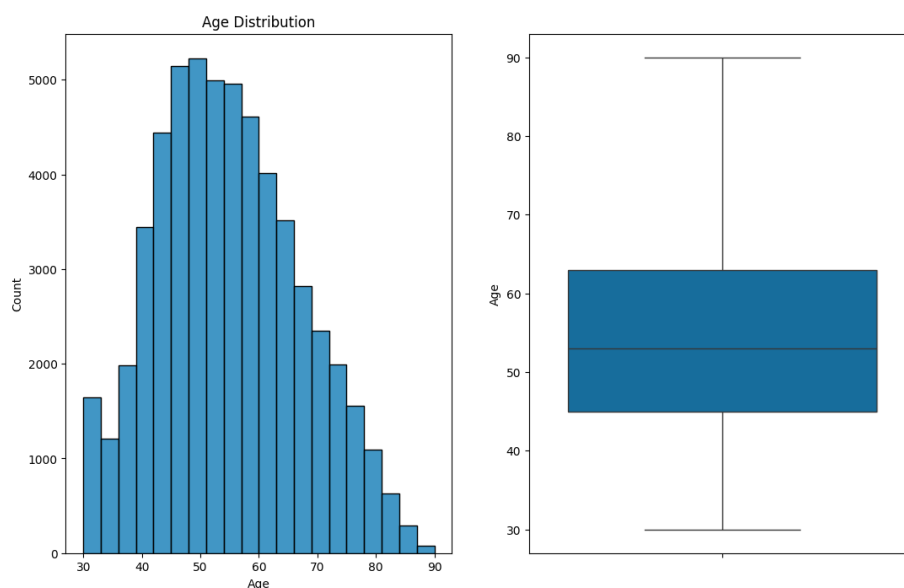


Figure 1: Age distribution and boxplot of customers

Customer age ranges from 30 to 90 years, with a mean of 54.3 and median of 53. Most customers fall between 40 and 65 years, indicating the company's primary customer base consists of middle-aged adults rather than younger shoppers.

Income Distribution

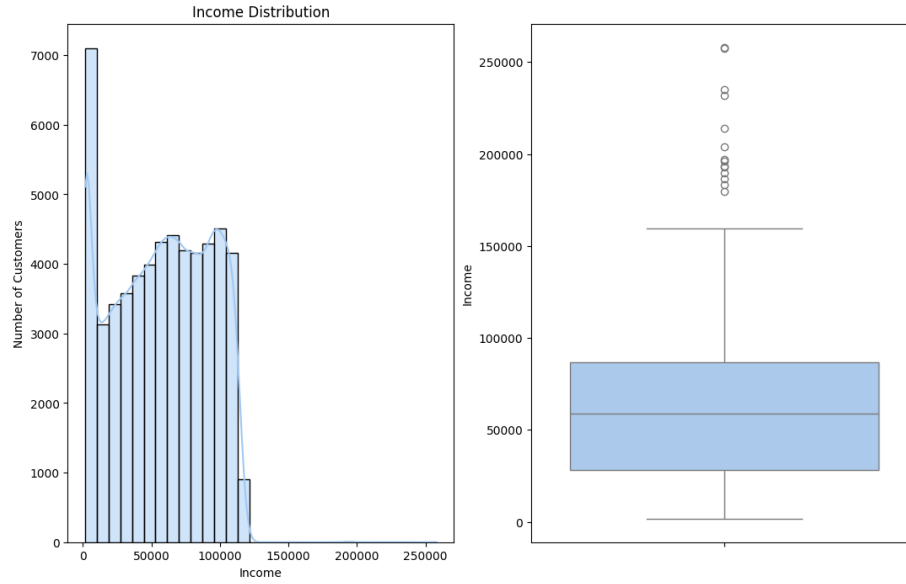


Figure 2: Income distribution and boxplot of customers

Income averages ₹57,252 (median ₹58,839) with a skewness value of approximately 0, indicating a fairly symmetrical distribution. The boxplot shows some high-income outliers; IQR analysis found 14 values above the upper bound (0.025% of records). These were investigated and found to correspond to genuine premium-income customers rather than data-entry errors, and were retained for analysis.

Education & Marital Status

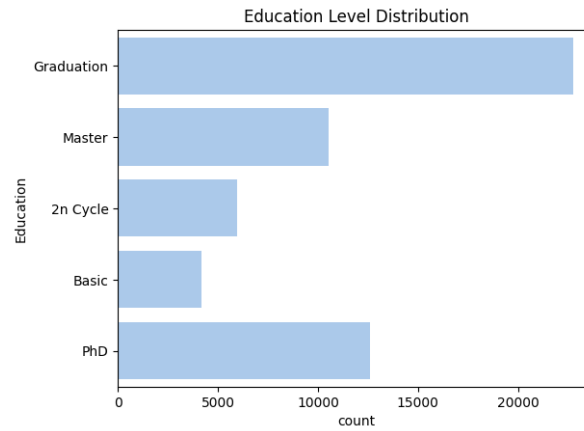


Figure 3: Education level distribution

Customers with a Graduation-level education dominate the customer base.

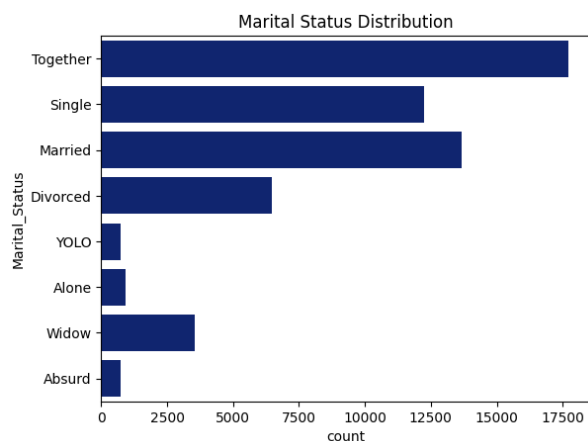


Figure 4: Marital status distribution

Married and Together customers form the two largest segments by count.

Country Distribution

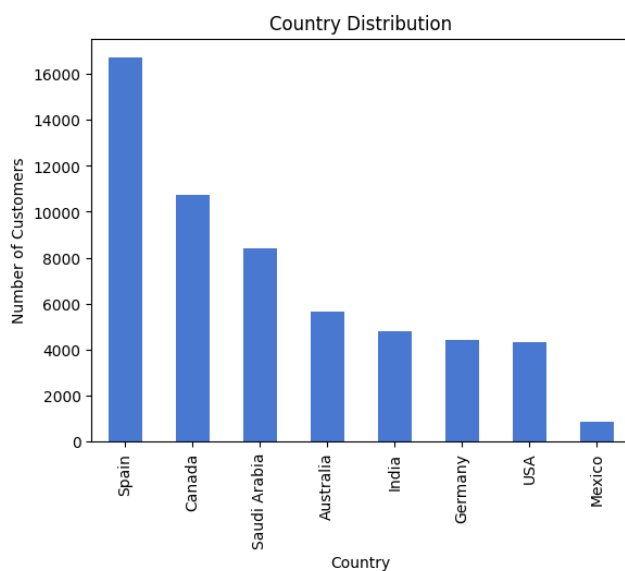


Figure 5: Customer distribution by country

Spain contributes the majority of customers, followed by Canada and Saudi Arabia, consistent with the revenue-by-country figures in the Executive Dashboard (Section 8).

4.2 Recency Analysis

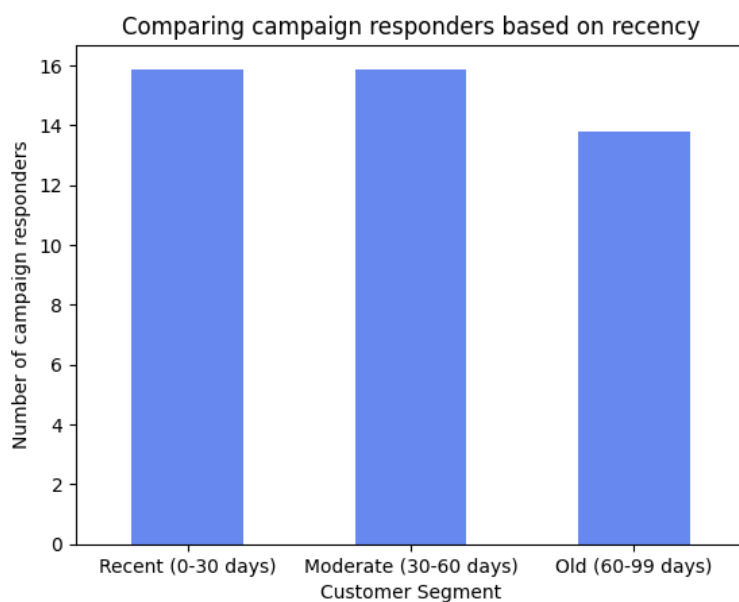


Figure 6: Campaign response rate by recency band

Recency Band	Response Rate	Avg. Total Spend
Recent (0–30 days)	15.87%	₹699.74
Moderate (30–60 days)	15.87%	₹670.34
Old (60–99 days)	13.77%	₹601.74

Recency values span 0–99 days with a large spike near 99, meaning a significant number of customers have not purchased in a long time. Customers who purchased more recently show a modestly higher response rate and higher average spend than those with older purchases.

4.3 Spending Analysis

Product-wise Spending

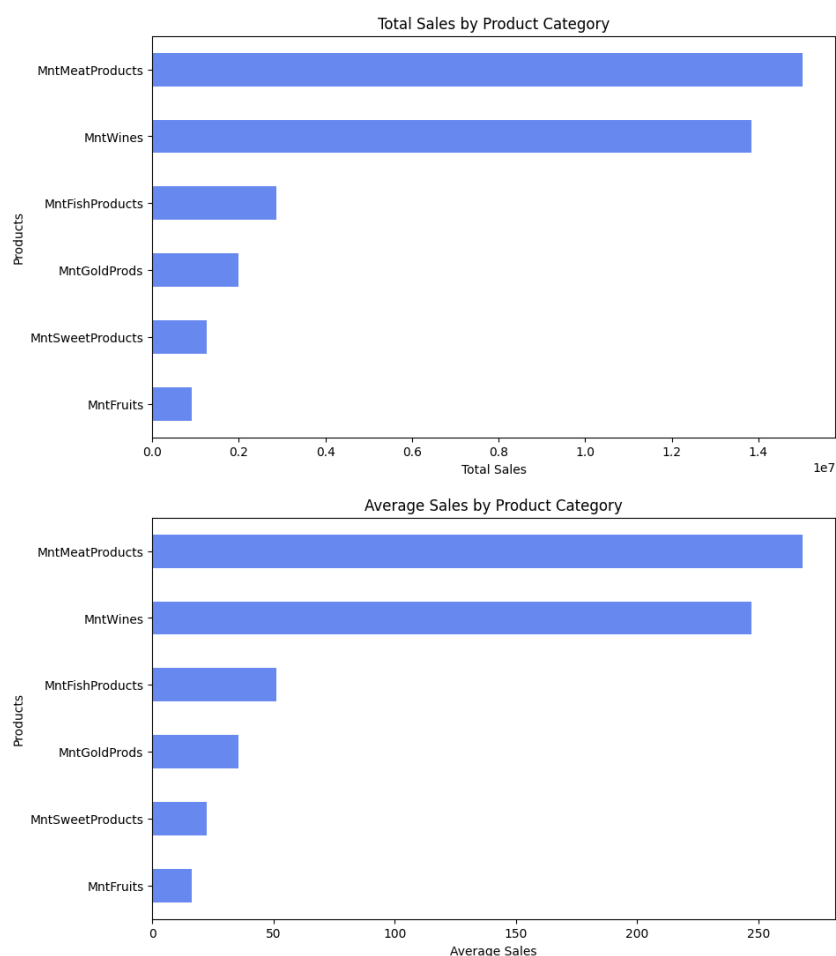


Figure 7: Total revenue and average spend by product category

Product Category	Total Revenue	Avg. Spend / Customer
Meat Products	₹15,024,465	₹268.29
Wines	₹13,830,963	₹246.98
Fish Products	₹2,861,473	₹51.10
Gold Products	₹1,981,595	₹35.39
Sweet Products	₹1,255,691	₹22.42
Fruits	₹904,549	₹16.15

Meat and Wine dominate total revenue, together contributing over 80% of product spend. This matches the Executive Dashboard's Product Revenue panel, where Meat (₹15.0M) and Wine (₹13.8M) lead all categories.

Spending vs. Income and Age

Income and Total Spend show a moderate positive correlation ($r = 0.573$) — higher-income customers generally spend more, though spending varies considerably among similarly-earning customers, suggesting other factors (age, family size, channel, campaign participation) also matter. Age and Total Spend show only a weak positive correlation ($r = 0.187$): age alone is not a strong predictor of spending.

Spending by Marital Status

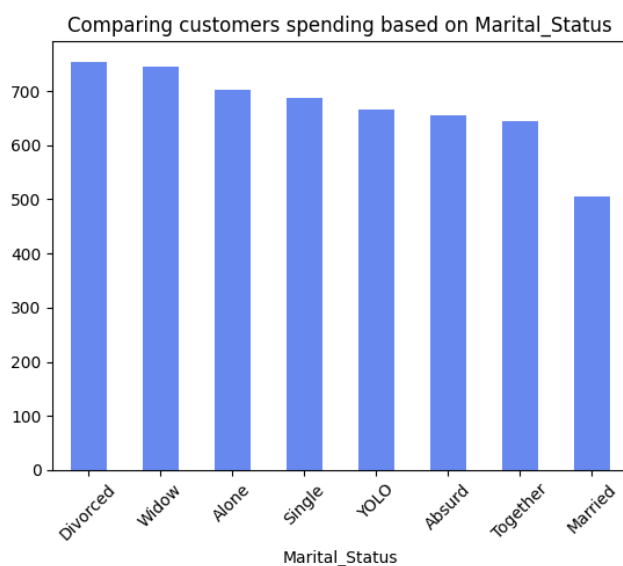


Figure 8: Average total spend by marital status

Divorced (₹753.91), Widow (₹745.05), and Alone (₹701.29) customers spend the most on average; Married customers spend the least (₹504.91) despite being the largest group by count — likely reflecting shared household budgets.

Spending by Number of Children

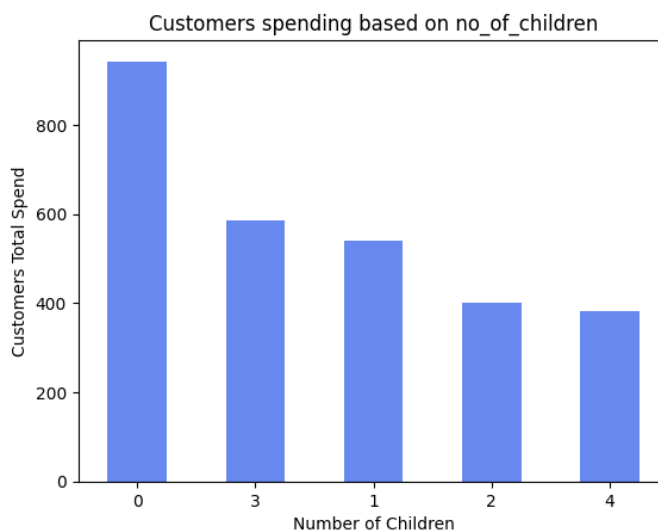


Figure 9: Average total spend by number of children

Customers with no children spend nearly double (₹942.70) that of customers with two or more children (as low as ₹400.71 for two children), showing family size is inversely related to average spend per customer.

Spending by Campaign Response

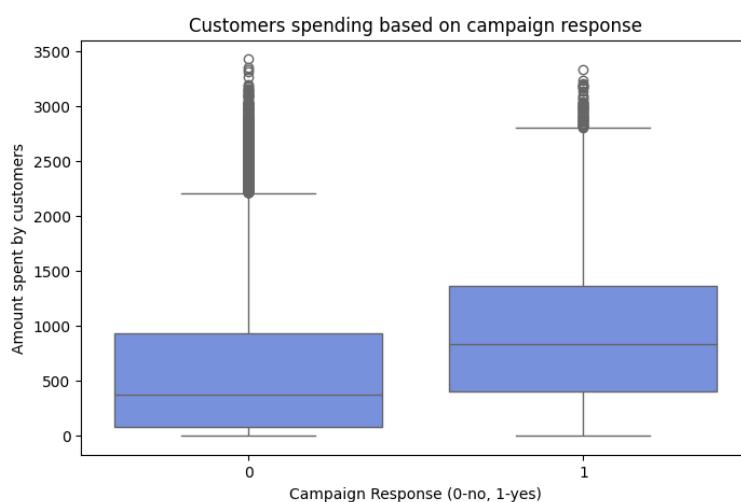


Figure 10: Total spend distribution by campaign response

Response	Mean Spend	Median Spend	Customers
Non-Responder (0)	₹588.57	₹375	47,735
Responder (1)	₹939.29	₹833	8,265

Customers who responded to campaigns have substantially higher average and median spend than non-responders, confirming that campaigns are more effective at attracting — or are more attractive to — already high-value customers. The presence of high-spending non-responders, however, shows room to improve targeting of valuable customers who haven't yet converted.

4.4 Channel Usage Analysis

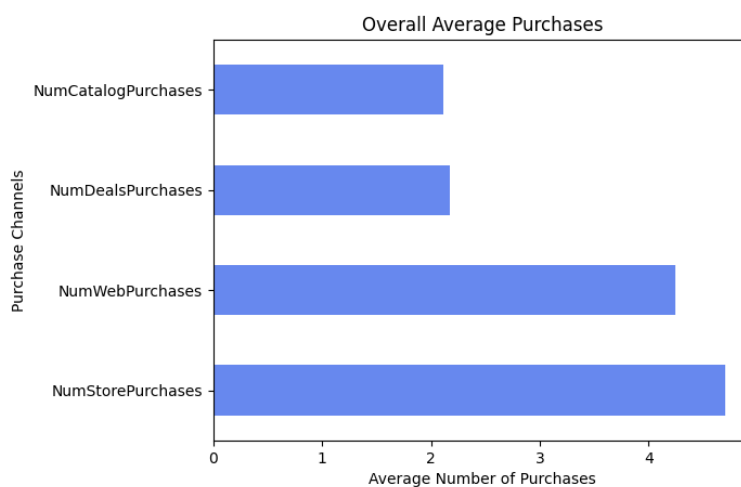


Figure 11: Average number of purchases by channel

Channel	Avg. Purchases / Customer	Skewness
Store	4.71	0.63 (moderate)
Web	4.25	1.10 (high)
Deals	2.17	1.59 (high)

Channel	Avg. Purchases / Customer	Skewness
Catalog	2.11	1.57 (high)

Store purchases are the most frequent and most evenly used channel across the customer base. Web, catalog, and deal purchases are all highly right-skewed, meaning most customers use these channels only lightly while a smaller subset of customers relies on them heavily — these channels are effectively dominated by specific customer segments rather than the general population.

4.5 Web Engagement Analysis

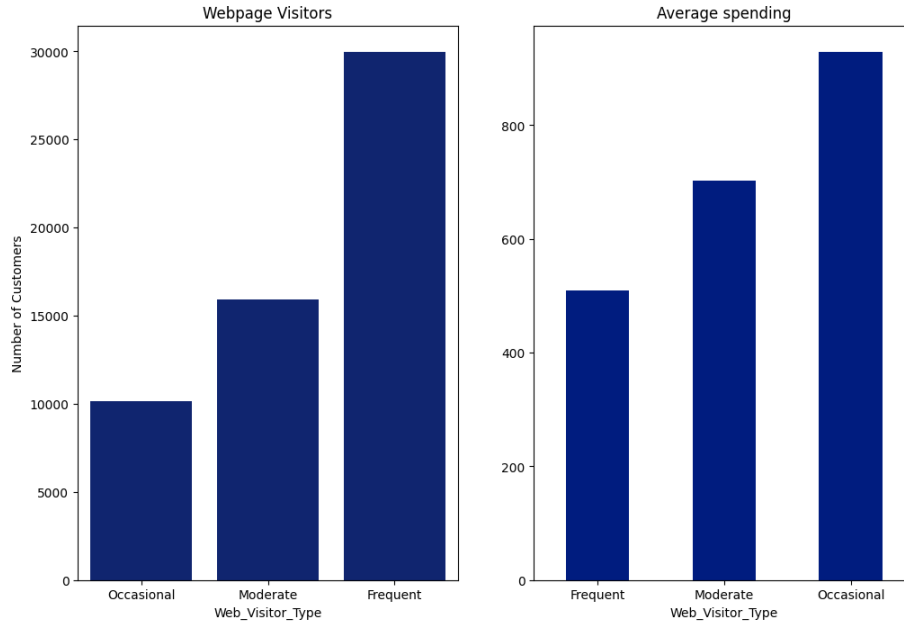


Figure 12: Web visitor type vs. spending

Web Visitor Type	Avg. Monthly Visits	Avg. Web Purchases	Avg. Income	Avg. Spend	Response Rate
Frequent (>5 visits)	7.20	3.91	₹48,627	₹510.08	12.07%
Moderate (3–5 visits)	4.06	4.49	₹62,405	₹701.88	16.01%
Occasional (≤2 visits)	0.90	4.87	₹74,687	₹929.17	20.75%

Counter-intuitively, customers who visit the website most frequently spend and respond the least — both spend ($r = -0.264$) and response ($r = -0.097$) show a weak negative correlation with monthly website visits. Occasional visitors have the highest income, spend the most, and convert at the highest rate, suggesting frequent browsing reflects window-shopping or unresolved friction in the online journey rather than purchase intent — a clear signal that online conversion, not just traffic, needs attention.

5. Campaign Analysis

5.1 Campaign Acceptance Overview

Campaign	Acceptance Rate
Campaign 1	13.44%
Campaign 2	1.44%
Campaign 3	6.24%
Campaign 4	5.68%
Campaign 5	4.57%
Latest Campaign (Response)	14.76%

Campaign 1 and the latest campaign (Response) performed far better than Campaigns 2–5, which struggled to break 7% acceptance. Campaign 2 in particular under-performed severely at 1.44%, suggesting either poor targeting or an unappealing offer for that run — worth a closer look before repeating a similar campaign design.

5.2 Response Rate by Demographics

By Age Group

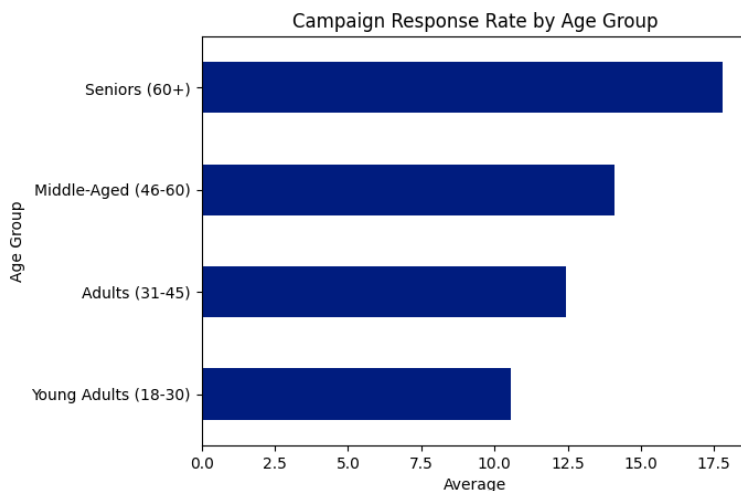


Figure 13: Campaign response rate by age group

Age Group	Response Rate
Young Adults (18–30)	10.57%
Adults (31–45)	12.43%
Middle-Aged (46–60)	14.11%
Seniors (60+)	17.80%

By Income Band

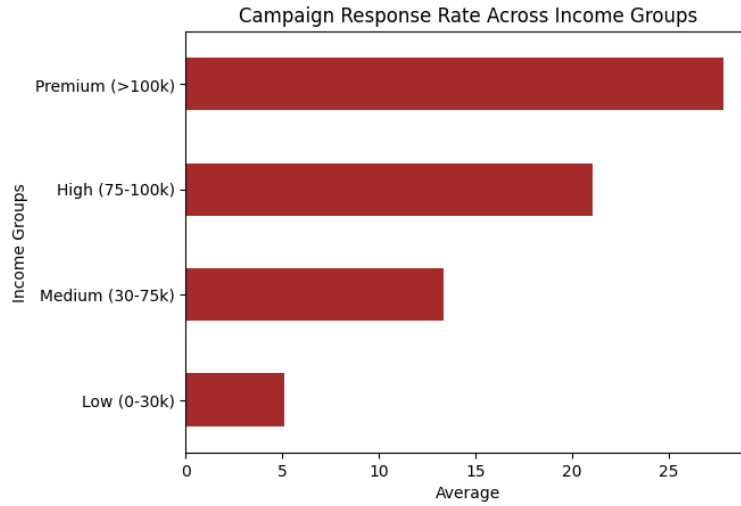


Figure 14: Campaign response rate by income band

Income Band	Response Rate
Low (₹0–30k)	5.07%
Medium (₹30k–75k)	13.32%
High (₹75k–100k)	21.05%
Premium (>₹100k)	27.81%

Response rate rises sharply and consistently with income — Premium-income customers respond over 5x more often than Low-income customers, making income the single clearest demographic lever for campaign targeting.

By Marital Status

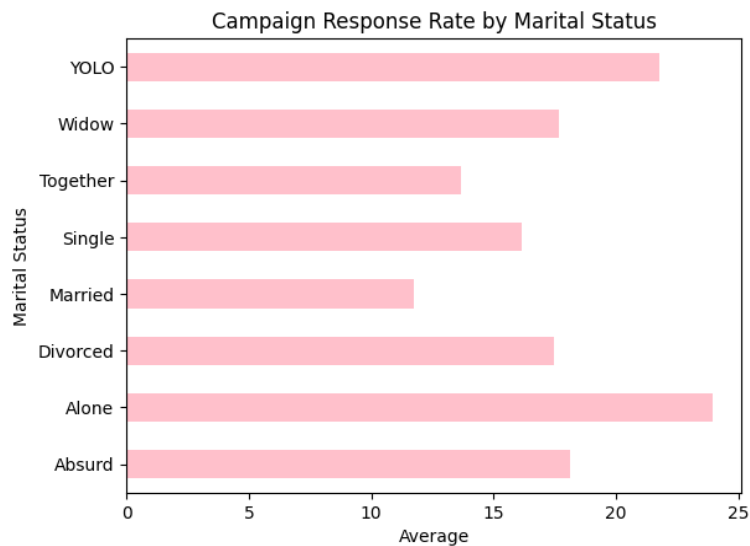


Figure 15: Campaign response rate by marital status

Marital Status	Response Rate
Alone	23.93%

Marital Status	Response Rate
YOLO	21.78%
Absurd	18.14%
Widow	17.68%
Divorced	17.48%
Single	16.17%
Together	13.65%
Married	11.72%

Single-person households (Alone, Widow, Divorced, Single) consistently out-respond partnered households (Married, Together), echoing the spending pattern seen by number of children — customers without a partner or dependents are both higher spenders and higher responders.

By Education Level

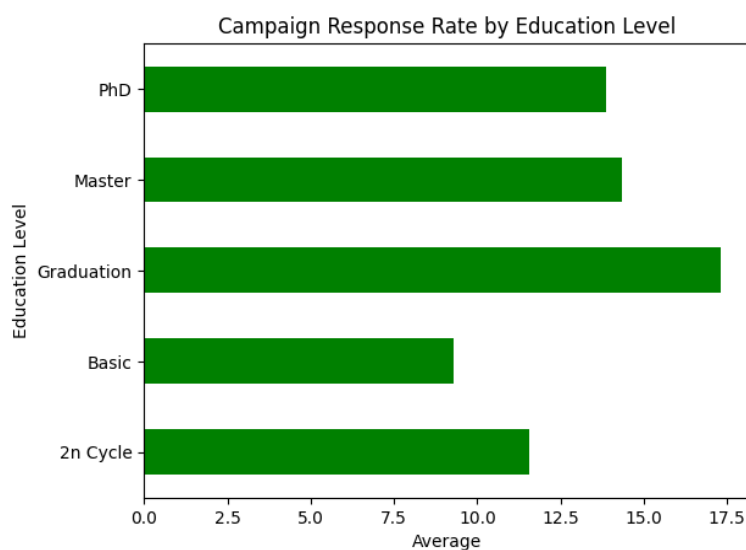


Figure 16: Campaign response rate by education level

Education	Response Rate
Graduation	17.30%
Master	14.35%
PhD	13.85%
2n Cycle	11.55%
Basic	9.28%

By Country

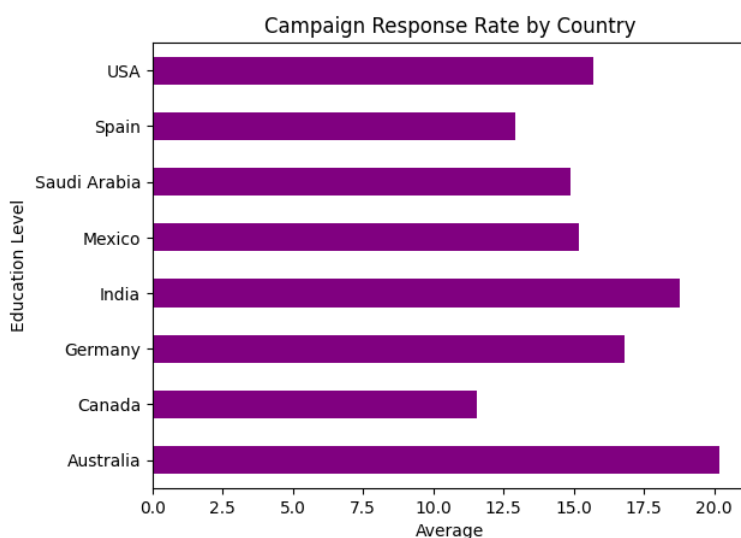


Figure 17: Campaign response rate by country

Country	Response Rate
Australia	20.19%
India	18.76%
Germany	16.81%
USA	15.70%
Mexico	15.20%
Saudi Arabia	14.90%
Spain	12.93%
Canada	11.57%

Spain generates the highest total revenue (₹9.31M, see Section 8) simply due to customer volume, but its response rate (12.93%) is below average — while Australia and India respond at notably higher rates despite lower customer counts. This gap between volume and responsiveness is a useful input for prioritizing future campaign spend by country.

5.3 Responder vs. Non-Responder Profile

Metric	Non-Responders	Responders
Avg. Age	54.02	56.13
Avg. Income	₹54,124	₹75,317
Avg. Total Spend	₹588.57	₹939.29
Avg. Children	0.94	0.69
Avg. Recency (days)	63.83	59.71
Avg. Web Visits/Month	5.27	—
Avg. Purchases	12.84	15.56

- Responders have higher average income and spend more overall than non-responders.
- Responders made a more recent purchase (lower recency).
- Responders visit the website less often but purchase more — suggesting they are decisive buyers, not browsers.
- Responders have fewer children on average, consistent with the single/couple spending pattern seen earlier.

6. Correlation Analysis

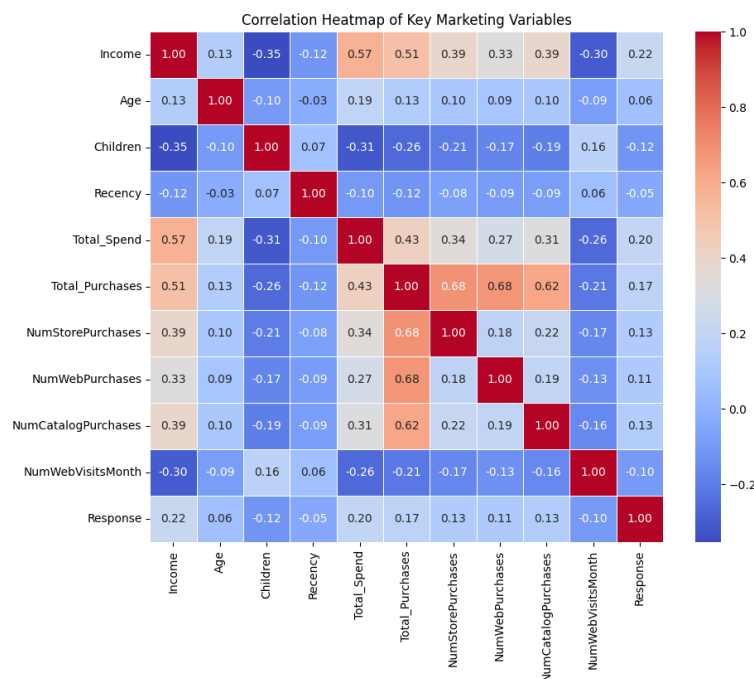


Figure 18: Correlation heatmap of key numeric variables

Income is one of the strongest drivers of customer value, showing positive relationships with both total spending ($r = 0.57$) and total purchases ($r = 0.51$). Customers with higher purchasing activity tend to spend more, while those with more children generally show lower income, spending, and purchase frequency. Frequent website visits do not correspond to higher spending — reinforcing the web-engagement finding in Section 4.5 and pointing to an online-conversion opportunity. Campaign response is moderately associated with income and spend, confirming that higher-value customers are more receptive to marketing efforts.

7. Customer Segmentation

Both a rule-based segmentation (per the project's business-defined rules) and behavioural groupings (income band, age band, family size, web engagement, high spenders) were implemented in Python and validated in SQL. Segment sizes shown here match the Customer Segments panel of the Executive Dashboard.

7.1 Rule-Based Segments (Business Rules)

Segment	Rule	Customer Count
Family Customer	Children > 0	38,000
High Web Engagement	NumWebVisitsMonth > 5	30,000
Campaign Responder	Response = 1	8,000
High Spender	Total_Spend > 90th percentile	6,000

7.2 Income-Based Segments

Income Band	Avg. Income	Avg. Spend	Response Rate
Low (₹0–30k)	₹12,304	₹194.19	5.07%
Medium (₹30–75k)	₹53,324	₹580.13	13.32%
High (₹75–100k)	₹87,743	₹927.32	21.05%
Premium (>₹100k)	₹107,465	₹1,228.29	27.81%

7.3 Age-Based Segments

Age Group	Avg. Income	Avg. Spend	Response Rate	Avg. Purchases
Young Adults (18–30)	₹49,256	₹462.30	10.57%	—
Adults (31–45)	₹52,833	₹508.68	12.43%	—
Middle-Aged (46–60)	₹55,190	₹611.57	14.11%	—
Seniors (60+)	₹64,231	₹796.94	17.80%	14.28

Customer value increases with age: Seniors (60+) are the most valuable segment, with the highest income, spend, purchase frequency, and response rate. Young Adults are the least engaged, suggesting future campaign spend may yield better returns by prioritizing middle-aged and senior customers over younger ones.

7.4 Family-Based Segments

Family Type	Avg. Income	Avg. Spend	Response Rate	Avg. Purchases
Single/Couple (0 children)	₹76,083	₹942.70	21.51%	15.58
Small Family (1 child)	₹50,986	₹539.76	12.29%	12.45
Large Family (2+ children)	₹42,871	₹409.31	10.10%	11.46

7.5 Web Engagement Segments

Web Visitor Type	Avg. Income	Avg. Spend	Response Rate	Avg. Purchases
Frequent (>5 visits)	₹48,627	₹510.08	12.07%	12.21

Web Visitor Type	Avg. Income	Avg. Spend	Response Rate	Avg. Purchases
Moderate (3–5 visits)	₹62,405	₹701.88	16.01%	13.92
Occasional (≤ 2 visits)	₹74,687	₹929.17	20.75%	—

7.6 High Spenders (Top 10% by Spend)

Segment	Avg. Income	Avg. Spend	Response Rate	Avg. Purchases
High Spender (>90th pct.)	₹87,842	₹1,972.20	26.04%	16.93
Regular Customer	₹53,854	₹492.38	13.51%	12.83

High Spenders respond to campaigns nearly twice as often as regular customers (26.04% vs. 13.51%) and purchase far more frequently — this segment alone is disproportionately valuable and should be a retention priority.

8. Under-Served Customer Segment

Under-served customers were defined per the project rules as those who: visit the website more than 5 times a month, spend below the overall median Total_Spend, and have not responded to any campaign (Response = 0). This segment represents customers who show clear interest but are not converting.

Metric	Overall Customer Base	Under-Served Segment
Count	56,000	16,765 (29.94% of base)
Avg. Income	₹57,252	₹32,309
Avg. Age	54.33	51.71
Avg. Total Spend	₹640.33	₹129.78
Avg. Web Visits/Month	5.17	7.24

Nearly 30% of the entire customer base falls into this under-served category — a substantial pool of high-intent, low-conversion customers. They are on average lower-income and slightly younger than the overall base, but visit the website considerably more often, indicating active interest that isn't translating into purchases or campaign engagement.

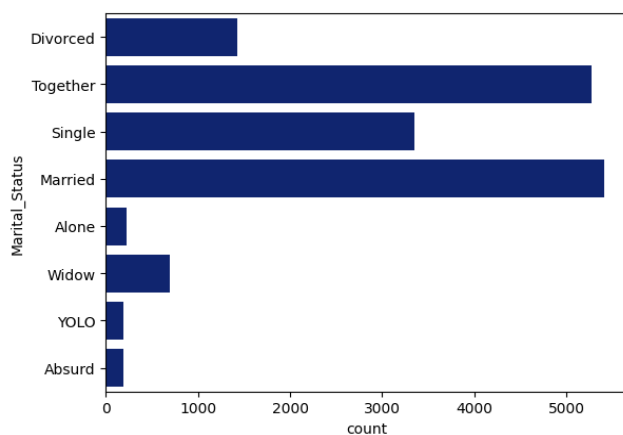


Figure 19: Under-served customers by marital status

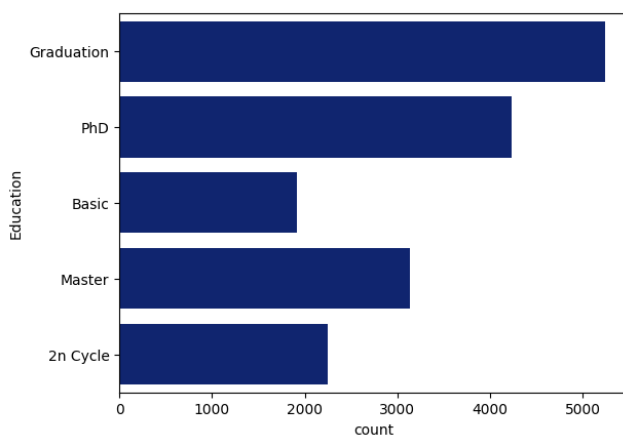


Figure 20: Under-served customers by education

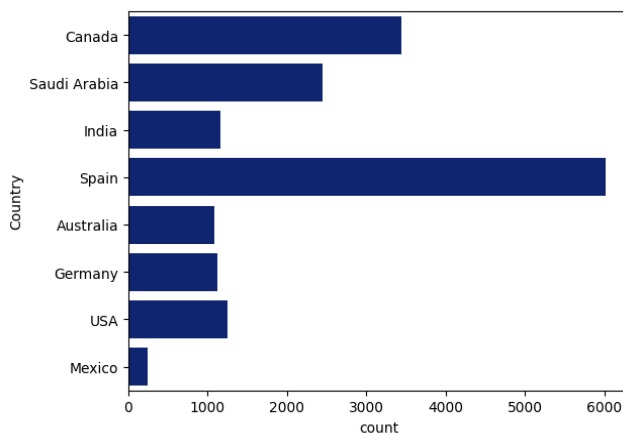


Figure 21: Under-served customers by country

Married and Together customers make up the majority of the under-served segment (over 60% combined), and Graduation/PhD-educated customers are over-represented relative to Basic education. Spain, Canada, and Saudi Arabia contribute the largest shares by country — closely tracking overall customer volume by country rather than indicating a specific geographic problem.

9. SQL Data Model & KPIs

A MySQL database (Marketing_Campaign) was created with a primary customers table (ID as primary key, appropriate data types for demographics, spend, channel, and campaign fields). Cleaned data from Python was loaded directly into this table.

9.1 Analytical Query Coverage

Comment-documented SQL scripts were written to compute: overall KPIs (total customers, average income, average spend, response rate); demographic breakdowns (age group, education, marital status, country); spending analysis by country, education, marital status, age group, and product category; campaign analysis (per-campaign acceptance, response by education/country/marital status/income band/age group, responder characteristics, product and channel preference of responders); channel analysis (average and total purchases per channel, web visits by response and income group); and segment-level summaries matching every rule-based and behavioural segment in Section 7.

9.2 Reusable Views

Seven SQL views were created to support the dashboard layer directly, avoiding repeated recomputation:

- vw_kpi_summary — total customers, revenue, average income, average spend, response rate
- vw_country_revenue — revenue and response rate by country
- vw_campaign_summary — acceptance rate per campaign (1–5) plus latest response
- vw_product_summary — total revenue by product category
- vw_channel_summary / vw_channel_totals — average and total purchases by channel
- vw_segment_summary — customer count, average spend, and response rate per rule-based segment

10. Dashboard Overview

The Power BI report has two pages — an Executive Dashboard for top-line KPIs and revenue/campaign performance, and a Customer Insights Dashboard for segment-level behaviour — both filterable by Country, Education, Marital Status, Age, and Income.

10.1 Executive Dashboard

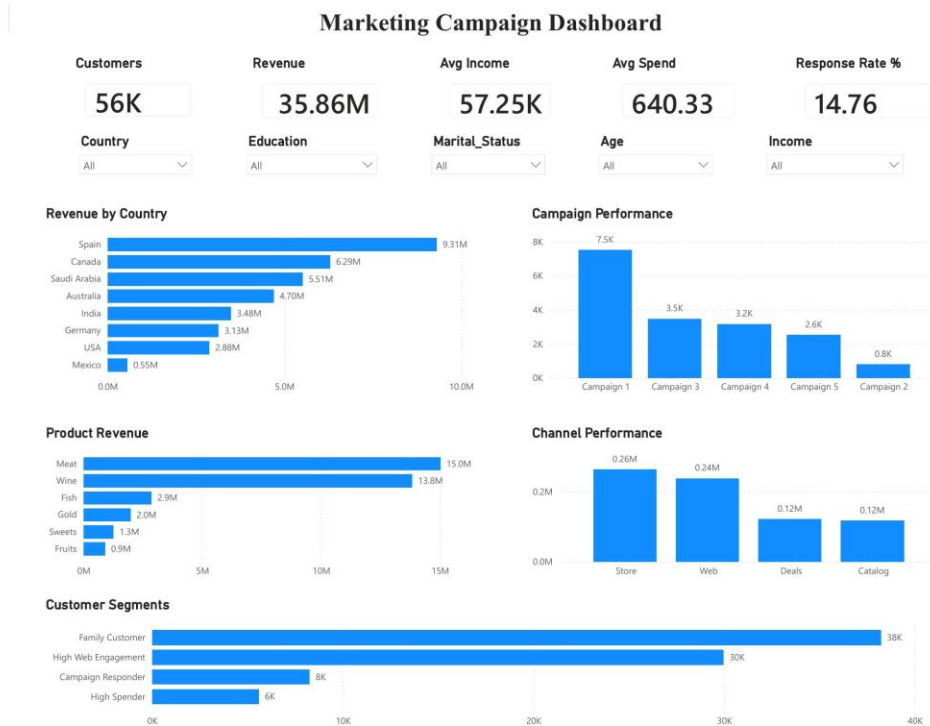


Figure 22: Executive Dashboard — KPIs, revenue by country, campaign performance, product revenue, channel performance, and customer segments

At a glance: 56,000 customers generated ₹35.86M in revenue, at an average income of ₹57.25K and average spend of ₹640.33, with a 14.76% overall response rate. Spain leads revenue by country (₹9.31M), Campaign 1 leads campaign performance (7.5K acceptances), Meat and Wine dominate product revenue, Store and Web are the top two channels, and Family Customer (38K) is the largest of the four highlighted rule-based segments.

10.2 Customer Insights Dashboard

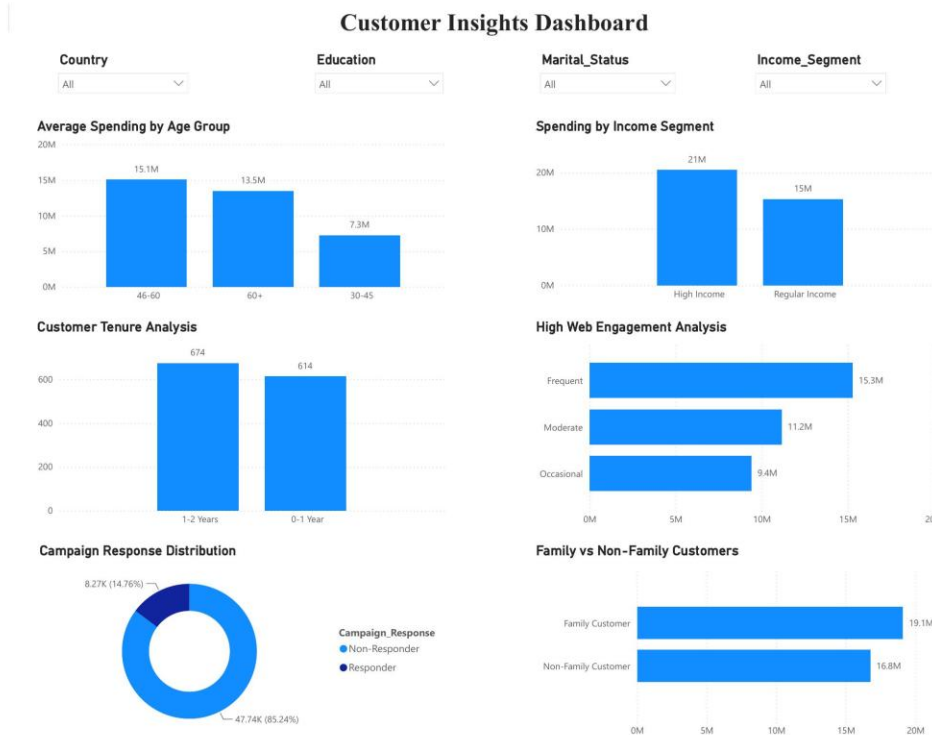


Figure 23: Customer Insights Dashboard — spending by age group and income segment, tenure, web engagement, response distribution, and family vs. non-family spend

This page confirms the age-based finding from Section 7.3: the 46–60 age group contributes the most total spend (₹15.1M), narrowly ahead of the 60+ group (₹13.5M), with 30–45 well behind (₹7.3M) — total spend here reflects both per-customer value and segment size. High Income customers contribute ₹21M in spend versus ₹15M for Regular Income. Frequent web visitors account for the largest share of spend by engagement tier (₹15.3M) simply due to segment size, even though — as shown in Section 4.5 — their per-customer spend is actually the lowest of the three tiers. The response distribution confirms the 14.76% overall rate (8.27K responders vs. 47.74K non-responders), and Family Customers out-spend Non-Family customers in aggregate (₹19.1M vs. ₹16.8M).

11. Key Findings Summary

- Income is the strongest single driver of both spending and campaign response — Premium-income customers respond over 5x more often than Low-income customers.
- Value increases with age and decreases with family size: Seniors (60+) and Single/Couple households are the most valuable and most responsive segments; Young Adults and Large Families are the least.
- Frequent website visitors are not the best customers — they spend and respond the least of the three engagement tiers, while Occasional visitors convert best. High website traffic should not be mistaken for high purchase intent.
- Meat and Wine drive over 80% of product revenue; Store remains the most consistently used purchase channel.
- Nearly 30% of customers (16,765) are 'under-served' — high website engagement, low spend, zero campaign response — representing the single largest untapped growth opportunity in the dataset.

- The High Spender segment (top 10%) responds to campaigns at nearly double the rate of regular customers and should be prioritized for retention, not just acquisition.

12. Recommendations

- 1. Retarget the under-served segment with personalized, low-friction offers.** The 16,765 high-visit, low-spend, zero-response customers are the largest single opportunity in the dataset. Use browsing-behaviour-based product recommendations, reminder emails/notifications, and first-purchase incentives to convert interest into transactions.
- 2. Prioritize Premium-income, Senior, and Single/Couple customers for future campaigns.** These segments consistently show the highest response rates (up to 27.8% for Premium income, 23.9% for customers living alone) and highest average spend — they offer the best expected return per campaign dollar.
- 3. Diagnose and fix the online conversion gap.** Frequent website visitors spend and respond the least of any engagement tier, despite the highest activity. Investigate site UX, checkout friction, and product discoverability rather than simply driving more traffic.
- 4. Re-evaluate or retire underperforming campaign formats.** Campaign 2 (1.44% acceptance) and Campaigns 3–5 (4.6%–6.2%) badly trail Campaign 1 and the latest campaign (13–15%). Review what made Campaign 1 and the most recent campaign more effective and apply those elements going forward.
- 5. Protect and grow the High Spender segment.** This top-10%-by-spend group responds at nearly double the average rate (26.0% vs. 13.5%) — a loyalty program or early-access offers could further increase their already-high purchase frequency (16.9 purchases on average).
- 6. Shift geographic campaign investment toward responsiveness, not just volume.** Spain drives the most revenue by sheer customer count but has a below-average response rate (12.9%); Australia (20.2%) and India (18.8%) convert far better per customer and may be under-invested relative to their potential.

13. Conclusion

This analysis combined Python-based EDA, a structured MySQL data model, and an interactive two-page Power BI dashboard to build a complete picture of customer value and campaign effectiveness for the retail company. The clearest pattern across every cut of the data — income band, age group, family size, and web engagement — is that value and responsiveness move together: higher-income, older, single/couple customers who visit the site less but buy more are consistently the company's best customers. The single biggest growth opportunity identified is the under-served segment of nearly 17,000 high-traffic, low-conversion customers, and the clearest efficiency opportunity is reallocating campaign investment toward the segments and geographies that respond best rather than simply those with the largest customer counts.